Gap2Idea: A Selective Verification Funnel for Limitations and Future Work

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Abstract

Scientific papers contain limitations and future-work statements that can support literature review and research planning, but extracting these statements requires both source attribution and control of processing cost. We study a three-stage funnel: structural localization, sentence detection, and selective large-language-model (LLM) verification. The implementation retains source sentences rather than generating gap descriptions. We evaluate saved experiments separately for full-article limitation detection, future-work recognition, and an auxiliary literature-gap task. On a 40-article randomized-trial evaluation stream containing 7,968 sentences and 205 labelled limitations, localization retains 203 positives while discarding 76.3% of sentences. A fine-tuned PubMedBERT detector then achieves precision 0.460, recall 0.878, and F_1 0.604. Selective verification increases precision to 0.580 and F_1 to 0.658, while reducing recall to 0.761. An estimated input-token comparison on these articles gives 282,846 tokens for full-article prompting versus 75,622 for selective verification, a 3.74-fold reduction; this estimate excludes detector computation and output tokens. Classical baselines, localization ablations, detector-development studies, and prevalence sensitivity analyses show that rankings depend on the input condition and target definition. We also document model-labelled reference construction and a human-validation protocol, without treating pending annotation as completed evaluation. Because prompts were refined using benchmark feedback, the LLM results are exploratory rather than untouched-test estimates. The study establishes a measured precision–recall and input-volume trade-off, without claiming general superiority over published systems or validation of downstream idea generation.

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1 Introduction

Research papers often tell their readers where a study falls short and what should be investigated next. In clinical research, self-acknowledged limitations help readers interpret findings in light of study design, data, and applicability [Kilicoglu et al., 2018]. In other disciplines, future-work statements identify extensions, unresolved questions, and further evaluations that authors consider worth pursuing [Zhu et al., 2019, Zhang et al., 2022]. We refer to these passages as *author-stated research gaps*: limitations and future-work directions expressed about the authors' own work. Their extraction provides a way to make this information searchable without requiring a model to invent a criticism or a research direction.

A collection of such statements can support several activities. A researcher may look for recurring restrictions in evaluation, identify settings that authors have left untested, or compare how studies describe future directions. A reviewer may use extracted statements to locate the authors' discussion of evidence and applicability. Corpus-level work already analyzes limitation reporting and future-work categories, and traces whether later publications engage with earlier research directions [Lan et al., 2024a, Hao et al., 2020, Suray et al., 2024]. These uses motivate a reliable extraction layer, but their interpretation remains conditional: an author's statement records a reported limitation or direction, not an independently established open problem. Later work may have addressed it, and the author may not have acknowledged every important weakness.

The first challenge is therefore definitional. A sentence criticizing prior research can be useful in a literature review yet lie outside the extraction of a study's own limitations. Similarly, a statement about how the current paper extends earlier work need not describe future research. Annotation schemes distinguish these functions in different ways. Argumentative zoning and scientific-concept annotation include categories related to gaps, failures, and future work [Liakata et al., 2010], while full-text resources annotate limitations and rhetorical roles in separate layers [Fisas et al., 2016]. Sci-Sentence provides explicit Research Gap, Limitation, and Extension categories for literature-review analysis [Bolaños et al., 2025]. Their names are related to our objective, but their definitions and attribution targets are not interchangeable. An extraction study must state which source claims count before comparing methods.

The second challenge is structural. Authors may place limitations in a dedicated section, within a discussion or conclusion, or alongside a particular methodological choice. Dedicated sections facilitate localization, but their surrounding text also contains results, mitigation, contribution claims, and descriptions of other studies. Restricting attention to terminal sections can remove irrelevant material cheaply, while also excluding mid-paper statements. PDF reading order, broken hyphenation, and sentence boundaries add a further source of error. A correct sentence classifier cannot recover a gold statement that the parser or localizer has already omitted. Evaluating localization and detection separately is therefore necessary to understand the recall of the complete system.

Prior extraction work provides useful precedents. Clinical limitation recognition uses headings, lexical cues, and supervised or semi-supervised classifiers [Kilicoglu et al., 2018]. Later work on randomized controlled trials (RCTs) separates limitation-sentence detection from classification into limitation types [Lan et al., 2024a]. Future-research recognition in JASIST articles

and future-work recognition in NLP similarly separate locating relevant sentences from grouping or classifying their content [Zhu et al., 2019, Zhang et al., 2022]. These studies motivate the use of inexpensive structural and lexical evidence before more flexible semantic classification. They do not imply that one detector transfers equally well between biomedical full articles, NLP conclusions, and literature-review sentences.

Large language models (LLMs) offer another route, but they introduce a distinct allocation problem. A full-article prompt presents many sentences unrelated to the extraction target, whereas a candidate-only prompt depends on the recall of the earlier stages. LLM-based work also combines source extraction with generation: LimGen uses author-written limitation sections as reference material, BAGELS evaluates limitation extraction and generation, and FutureGen extracts and then generates future-work text [Faizullah et al., 2024, Al Azher et al., 2025a,b]. Critical-limitation and unstated-limitation studies instead ask models to identify weaknesses that may not have been explicitly acknowledged [Xu et al., 2025, Mishra et al., 2026]. The ability to produce a plausible criticism does not establish the ability to identify an existing author-stated sentence. Our objective is the latter, with source attribution retained.

We investigate a *selective verification funnel* that divides this extraction task into three stages. Stage A selects candidate regions using document structure. Stage B combines sentence detection, with lexical cues in the hybrid configuration. Stage C optionally asks an LLM to verify the surviving candidates against the task's attribution policy. Each stage selects existing parsed sentences rather than rewriting them. This design makes the consequences of compression visible: we can measure how many labelled positives remain after localization, how many candidates the detector passes on, and which true and false positives the verifier removes.

The present experiments examine this mechanism under several distinct input conditions. The principal analysis uses reconstructed full-article SAL records from biomedical RCT reports, with complete saved detector predictions and verifier decisions. Separate studies concern curated future-work sentences in NLP, a literature-gap classification task, and a small arXiv development reference. Development studies further compare localizer variants, lexical baselines, class balancing, and alternative reference sets. A documented model-labelled corpus and its implemented human-validation workflow are distinguished from completed human evaluation. We also compare estimated LLM input and a recorded small full-article prompting baseline. Dataset-specific labels, prevalence, and processing units are kept explicit because pooling them would obscure what each experiment measures. These experiments concern extraction and its processing requirements; corpus-level characterization is a separate empirical objective.

Research questions. We ask three questions for the evidence reported in this paper:

- **RQ1 (Stage contribution).** How do structural localization, sentence detection, and selective LLM verification change limitation-detection precision, recall, and F_1 on a full-article stream?
- **RQ2 (Detector and reference dependence).** How does the observed trade-off vary across detector families, annotation targets, and reference construction, including author-stated future work and literature-gap recognition?

- **RQ3 (Input efficiency).** How much LLM input is avoided by selecting candidate sentences, and what effectiveness is recorded in a same-model full-article comparison?

Contributions. The study makes three contributions:

1. An extractive pipeline with explicit interfaces for localization, detection, and verification, together with the rationale for allocating structural, lexical, learned, and LLM processing to different stages.

2. A full-article stage analysis recomputed from saved sentence-level predictions and verifier decisions, accompanied by detector comparisons and separate auxiliary-task results that make the input condition and attribution target visible, alongside development ablations and reference-sensitivity analysis.

3. An input-token accounting study and a small same-model full-article comparison, reporting effectiveness and processing-volume trade-offs separately from unmeasured total billing or throughput.

Prompt development used benchmark feedback, so the verifier results are exploratory. The claim is a measured trade-off for the examined configurations, rather than general superiority of LLMs, independent validation across all disciplines, or demonstrated idea-generation quality.

Paper structure. Section 2 develops the literature comparison, including the chronological study-by-study table. Section 3 traces data acquisition, preparation, extraction, scoring, and the analyses used to answer each RQ. Section 4 reports the verified and aggregate results with their evidence status. Section 5 interprets the mechanisms and validity limits, and Section 6 summarizes the answers and the next evaluation steps.

2 Related Work

Our study connects the recognition of author-stated limitations and future work with rhetorical sentence classification, LLM-supported scholarly-text processing, and analysis of reported research directions. These areas share input documents and some methods, but their output claims differ. We review that relationship before positioning the current extraction evaluation.

2.1 Rhetorical annotation and scientific sentence classification

Argumentative zoning provides an early account of sentences according to their role in scientific argument [Teufel and Moens, 2002]. Liakata et al. [2010] describe AZ-II alongside Core Scientific Concepts (CoreSC), illustrating two complementary views of scholarly content. Roles such as future work, weaknesses in prior research, and failures of an author's approach can all help locate research-relevant statements, but they do not constitute a single uniform author-gap label. The attribution target and the distinction between a statement's rhetorical role and its semantic content remain part of the annotation definition.

Full-text annotation is also an established setting. The Dr. Inventor corpus contains 40 Computer Graphics articles with multiple annotation layers, including argumentative roles and

statements about advantages or disadvantages [Fisas et al., 2016]. The presence of limitation and future-work information in separate layers is relevant to our task: it shows why a joint extraction interface can draw on earlier annotation work without claiming to introduce the first full-text resource containing both kinds of statement.

Abstract classification resources offer a different input condition. PubMed 200k RCT provides biomedical abstract roles, while CSAbstruct supports classification of computer-science abstract sentences [Dernoncourt and Lee, 2017, Cohan et al., 2019]. MAZEA and Emerald 110k broaden the available rhetorical annotation schemes [Dayrell et al., 2012, Stead et al., 2019]. Such resources support document understanding, but an abstract role such as conclusion or objective does not establish that a sentence expresses an author-stated limitation. Treating every non-gap-specific role as a reliable negative would also require checking whether gap statements can occur inside that role.

Context matters for classification because a short sentence may inherit its subject or purpose from nearby text. Jin and Szolovits [2018] use a hierarchical sequence model to incorporate surrounding sentences in medical abstracts. Brack et al. [2024] examine cross-domain multi-task learning for scientific sentence classification, and Lan et al. [2024b] study multi-label classification with LLMs. These studies motivate explicit treatment of context, but they do not establish that complete-sequence modeling is always appropriate for sparse gap labels in full papers. Zhang and Zhang [2024] instead investigate context-enhanced problem/method sentence extraction, an adjacent task in which local context helps interpret a target sentence. Our verifier uses neighboring source sentences for attribution without replacing the task by whole-document rhetorical classification.

2.2 Author-stated limitation recognition

Kilicoglu et al. [2018] recognize self-acknowledged limitations in clinical research using rules and learned approaches. Their heading and lexical evidence provides a direct precedent for our localization and cue stages. The important relationship is mechanistic: document structure reduces the search space, while sentence-level evidence distinguishes limitation claims from surrounding exposition. This is more specific than treating all methods that process scientific text as equivalent extraction baselines.

Lan et al. [2024a] extend clinical limitation work to 200 annotated RCT publications with 15 categories and 24 subcategories. Their reported binary detector F_1 of 0.821 differs from the limitation-type classification result. Their at-scale application also characterizes limitation reporting. The dataset is useful to us because it supplies author-gap labels and non-positive sentences in an article-based setting. However, a reported candidate-pool recognition score cannot be compared directly with a reconstructed full-stream score: section filtering, label prevalence, training choices, and the recall denominator must first be aligned.

LimGen contributes author-written limitation sections from 4,068 ACL papers for suggestive limitation generation [Faizullah et al., 2024]. These sections provide useful reference and training material, but section membership is a form of supervision rather than proof that every sentence is a semantic limitation. The distinction matters when a precision filter rejects a method description, result statement, or checklist item occurring inside a Limitations section. Nor is the

original generation task an official binary-detection baseline simply because its references can be transformed into sentence labels.

BAGELS combines source parsing and LLM processing with limitation generation across NLP, ML, and biology [Al Azher et al., 2025a]. Its extraction evaluation reports different performance for ACL and NeurIPS samples, which the authors discuss in relation to explicit versus scattered limitation reporting. That comparison motivates evaluating structural accessibility, but does not isolate section placement from every other corpus difference. Reviewer evidence is another important distinction: author statements and reviewer-identified limitations are related sources of critique, not the same attribution target. We preserve that distinction in comparisons with our authors'-own extraction task.

2.3 Future-work recognition and characterization

Zhu et al. [2019] study future-research sentences in full JASIST articles using BERT-based recognition and subsequent clustering. Their work links sentence extraction to thematic analysis: retrieval locates source statements, whereas clustering characterizes what those statements concern. The two stages therefore answer different questions, and a useful thematic grouping does not independently verify that a stated research direction remains open.

ACL FWS-RC introduces an annotated future-work corpus and taxonomy [Hao et al., 2020]. The later recognition and classification study of Zhang et al. [2022] compares machine-learning methods and reports macro-$F_1 = 0.9073$ for recognition and weighted $F_1 = 0.7263$ for typing. These numbers must remain attached to their own tasks and evaluation protocols. The resource is particularly relevant to our future-work branch, while its curated sentence input makes it a different test from parsing and scanning a complete article. We accordingly report the FWS breadth experiment as sentence recognition rather than general full-PDF extraction.

Suray et al. [2024] analyze future-work statements in usable-security and privacy research and investigate their relationship with citing publications. This extends the purpose of extraction from locating sentences to examining how research agendas are articulated and subsequently used. It also makes interpretation cautious: a future-work statement can be vague, and later research can engage with a direction without repeating its wording. Such tracking would require additional corpus and citation evidence beyond the stage-level recognition experiments reported here.

2.4 LLM extraction, inferred gaps, and generated critique

LLMs can serve several roles in scholarly-text analysis. An extractor selects source passages; a verifier judges a candidate's fit to a definition; a generator proposes statements that need not appear in the source. BAGELS and FutureGen include both extraction and generation workflows [Al Azher et al., 2025a,b]. FutureGen's retrieval-augmented generation of future-work text is a downstream use of evidence, while our Stage C returns only a decision about an existing candidate. The shared use of an LLM does not make their outputs or evaluation measures interchangeable.

GAPMAP distinguishes explicitly expressed knowledge gaps from gaps inferred through context [Salem et al., 2025]. Its evaluation includes multiple input units, from paragraphs and

sections to a small full-text pilot. This is relevant to the document-level interpretation of gap claims, while also showing why dataset coverage must be separated from processing unit. A collection of paragraph-level cases cannot by itself establish a full-article extractor's localization performance, and inferred missing knowledge need not correspond to one quoted sentence from the authors.

LimitGen is a separate benchmark from LimGen. It evaluates critical-limitation identification on synthetic and reviewer-based AI research references [Xu et al., 2025]. Tree-of-Concerns addresses unstated limitations using hierarchical multi-agent critique [Mishra et al., 2026]. Both concern scientific weaknesses that a system may identify without the authors acknowledging them. They therefore help delimit our objective and possible downstream extensions rather than providing an automatically comparable extraction score. Retaining the original sentence can support provenance, but it does not establish the substantive validity or novelty of a generated criticism or research idea.

2.5 Comparison dimensions and the research question addressed

Table 1 compares the selected studies in chronological order by the year of the cited version. It separates evaluated domain, target statement, origin of the reference, method, dataset coverage, and processing unit. The gap-distribution column describes characterization of extracted or annotated statements, not hypothesis tests of classifier performance. The final column explains the connection of each study to our extraction design or research questions. Dataset coverage denotes the evaluated resource rather than a claim about where the method can be applied universally.

Several comparison decisions follow. Authors' own limitations and criticisms of prior work require different attribution policies. Binary recognition and fine-grained typing require different metrics and denominators. Full-article inputs and curated sentences impose different localization burdens. Author-written, reviewer-written, and generated references have different evidential origins. Finally, extraction effectiveness, corpus characterization, and research-idea usefulness are separate outcomes. These distinctions motivate our dataset-specific tables and the separation of input accounting from total system cost.

The present contribution concerns the allocation and evaluation of processing within an extractive funnel. Structural filtering, lexical rules, learned classifiers, and LLM refinement all have prior precedents. We examine their measured combination through a common full-article stage comparison, auxiliary target-dependent recognition results, and input-volume analysis. The resulting comparison positions selective verification as an allocation strategy whose effectiveness must be evaluated under an explicit target and input condition.

Table 1: Selected related work in ascending year order for the cited version. The relevance column explains each connection to the extraction task and research questions. Gap-distribution analysis excludes tests of classifier performance. The final row concerns the evidence reported here.

Study/year	**Evaluated domain**	**Statement type**	**Statement origin**	**Approach**	**Dataset coverage**	**Processing unit**	**Gap-distribution analysis**	**Why related to our study**
2016: Dr. Inventor corpus [Fisas et al., 2016]	Computer Graphics	Limitations; future work	Author-stated; separate annotation layers	Human multi-layer annotation	40 full articles	Sentences in full articles	Descriptive label distributions	Precedent for full-text annotations containing both statement types in distinct layers.
2018: Limitation recognition [Kilicoglu et al., 2018]	Clinical research	Limitations	Author-stated	Rules; supervised and semi-supervised classification	1,197 source articles	Sentences	Not investigated	Direct precedent for heading/keyword localization and learned author-limitation recognition (RQ1).
2019: Future-research recognition [Zhu et al., 2019]	Information science; JASIST	Future work	Author-stated	BERT classification; clustering	JASIST training and application collections	Sentences in full articles	Descriptive cluster analysis	Connects full-text future-work recognition to subsequent thematic analysis (RQ2).
2020: ACL FWS-RC [Hao et al., 2020]	NLP; ACL papers	Future work	Author-stated	Manual annotation; taxonomy	4,024 papers; 3,067 future-work sentences	Sentences	Descriptive categories and yearly trends	Provides future-work labels/taxonomy; distinguishes recognition from characterization.
2022: FWS recognition/typing [Zhang et al., 2022]	NLP; academic articles	Future work	Author-stated	Lexical/neural recognition; type classification	Annotated FWS recognition/typing corpora	Sentences	Keyword/content and later-research similarity analysis	Direct recognition comparator; positive/macro/weighted F1 and sample protocols must align.
2024: LimGen [Faizullah et al., 2024]	NLP; ACL Anthology	Limitations	Generated; author-stated references	Dataset; LLM generation	4,068 papers	Paper	No corpus-level inferential analysis	Source of author-written limitation references; generation and section supervision differ from detection.

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Table 1 (continued)

Study/year	Evaluated domain	Statement type	Statement origin	Approach	Dataset coverage	Processing unit	Gap-distribution analysis	Why related to our study
2024: Limitation categorization [Lan et al., 2024a]	Biomedical RCT publications	Limitations	Author-stated	PubMedBERT detection and type classification	200 annotated articles; about 12k in application	Sentences	Corpus distributions; annotated-sample confidence intervals	Provides the principal author-limitation target; distinguishes detection, typing, and corpus analysis.
2024: LLM-SSC [Lan et al., 2024b]	Biomedicine; computer science; physical/chemical sciences	Rhetorical roles	Author-written text; role labels	LLM prompting and fine-tuning	Multiple abstract datasets	Abstract sequence	Not investigated	Context-aware LLM sentence labeling is adjacent to candidate verification, with different labels.
2024: Problem/method extraction [Zhang and Zhang, 2024]	NLP	Problem; method	Author-written text; sentence labels	Context-enhanced transformer	305 abstracts; 74 full papers	Target sentence with adjacent context	Not investigated	Motivates local context for sparse target sentences; problem/method labels are not gap labels.
2024: Cross-domain SSC [Brack et al., 2024]	Multiple scientific domains	Rhetorical roles	Author text; rhetorical labels	Hierarchical cross-domain multi-task classification	Multiple abstract and full-paper datasets	Sentence sequence	Role and transfer analysis	Motivates detector/context alternatives and explicit transfer conditions (RQ2).
2024: Future work at SOUPS [Suray et al., 2024]	Usable security/privacy	Future work	Author-stated	Source-statement and citing-publication analysis	SOUPS proceedings and citing publications	Statement; citing paper	Characterization and subsequent engagement	Motivates later corpus use; recognition alone does not establish a direction remains open.
2025: BAGELS [Al Azher et al., 2025a]	NLP; ML; biology	Limitations	Author-stated; reviewer-identified; generated	Rules and parsing; LLM extraction and generation	ACL, NeurIPS, and PeerJ collections	Paper	Corpus-level analysis proposed as future work	Closest extraction/generation precedent; motivates source grounding and section-aware evaluation.

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Table 1 (continued)

Study/year	Evaluated domain	Statement type	Statement origin	Approach	Dataset coverage	Processing unit	Gap-distribution analysis	Why related to our study
2025: FutureGen [Al Azher et al., 2025b]	NLP; ML	Future work	Author-stated; reviewer directions; generated	LLM extraction; RAG generation	Several thousand ACL/NeurIPS papers	Paper	No corpus-level inferential analysis	Future-work extraction feeds generation; motivates selective input while keeping the tasks distinct.
2025: GAPMAP [Salem et al., 2025]	Biomedicine; biodiversity; multi-domain pilot	Knowledge gaps	Explicit; context-inferred	Dataset; LLM gap detection and inference	About 1,500 mixed-unit inputs; small full-text pilot	Paragraph; section; paper	No corpus-level inferential analysis	Clarifies explicit versus inferred gaps and the importance of the input unit.
2025: LimitGen [Xu et al., 2025]	AI; NLP and ICLR papers	Limitations	Synthetic; reviewer-identified	Benchmark; LLM and agent evaluation	Synthetic and reviewer-based benchmark cases	Paper	Descriptive aspect distribution	Defines an adjacent critical-limitation task; reviewer/synthetic targets differ from quoted author claims.
2025: Sci-Sentence [Bolaños et al., 2025]	Literature reviews across disciplines	Gap; limitation; other roles	Human rhetorical labels; augmented training text	Seven-class models; LLM/encoder comparisons	700 human-labelled sentences; augmented variant	Sentence	Rhetorical-class/error analysis	Auxiliary target includes prior-work criticism; requires label/prompt alignment (RQ2).
2026: Tree-of-Concerns [Mishra et al., 2026]	ML; NLP; computer vision	Limitations	Inferred; reviewer/citation references	Hierarchical multi-agent critique	ToC-Bench: 414 papers; 1,905 limitations	Paper	No corpus-level inferential analysis	Addresses unstated critique rather than author-stated extraction; bounds downstream claims.
This study	Biomedical full articles; NLP future work; auxiliary literature gaps	Author limitations/future work; separate auxiliary gap task	Existing author text; dataset-specific labels	Structural localization; detectors; selective LLM verification	40-article principal SAL stream; separate auxiliary subsets	Source sentence; article-level resampling	Not reported in the present extraction evaluation	Measures stage contribution (RQ1), target dependence (RQ2), and estimated input efficiency (RQ3).

3 Methodology

The funnel separates three decisions: which source regions to inspect, which sentences to retain as candidates, and which candidates an LLM accepts. We describe the reusable extraction runner and identify the configurations used in each experiment. The biomedical experiment substitutes in-domain binary detectors for the runner's three-class hybrid head; it is therefore a study of the same stage decomposition, not evidence that one unchanged classifier was deployed across all datasets.

3.1 Task definition

The principal task returns sentences expressing a limitation of the authors' own study or a future-work direction associated with that study. A limitation may concern study design, data, evaluation, assumptions, or applicability. A future-work statement proposes subsequent research rather than merely describing a contribution already made. The system does not determine whether an author-stated problem is still open in the wider literature.

Let a parsed article be $D = (s_1, \ldots, s_n)$. Stages produce nested index sets

$$I_C \subseteq I_B \subseteq I_A \subseteq \{1, \ldots, n\}. \tag{1}$$

An output contains the retained source sentence, paper identifier, predicted type, section label, acceptance source, confidence value, and local context. Text is selected rather than rewritten. This guarantees extraction from the *parsed representation*, not perfect correspondence to PDF typography or correct attribution. Parsing and classification errors remain possible. Rule confidence is a fixed value and classifier confidence is a model probability; neither is treated as a calibrated measure of factual correctness.

The Sci-Sentence auxiliary experiment uses a different positive definition: Limitation or Research Gap, including statements about prior work. Its target-matched verifier permits that attribution. These results are reported separately from the principal task. Threats to validity can trigger a target-section rule, but no independently evaluated third output class is claimed.

3.2 Evaluation data and evidence provenance

Table 2 lists the evaluated subsets rather than conflating them with the size of each upstream resource. The principal SAL artifact contains 40 articles, 7,968 ordered sentences, and 205 limitation labels. Its records include article identifiers, recovered section tags, gold labels, and predictions from eight detectors. Section tags were recovered through PMC BioC in the original experiment. This reconstruction is held fixed in our offline analysis; its quality is not established by the sentence-classification metrics.

The SAL records are a reconstructed evaluation stream, not a new annotation effort. We do not assume that an unlabelled sentence is certainly free of a semantically valid limitation, nor replace source labels on the basis of an LLM's disagreement. The FWS breadth script removes exact training-sentence matches and samples papers with seed 0 until the selected pool reaches about 1,500 sentences. This sentence exclusion is not proof of complete paper-disjointness. Sci-Sentence uses its supplied 140-sentence test partition. Raw upstream files and

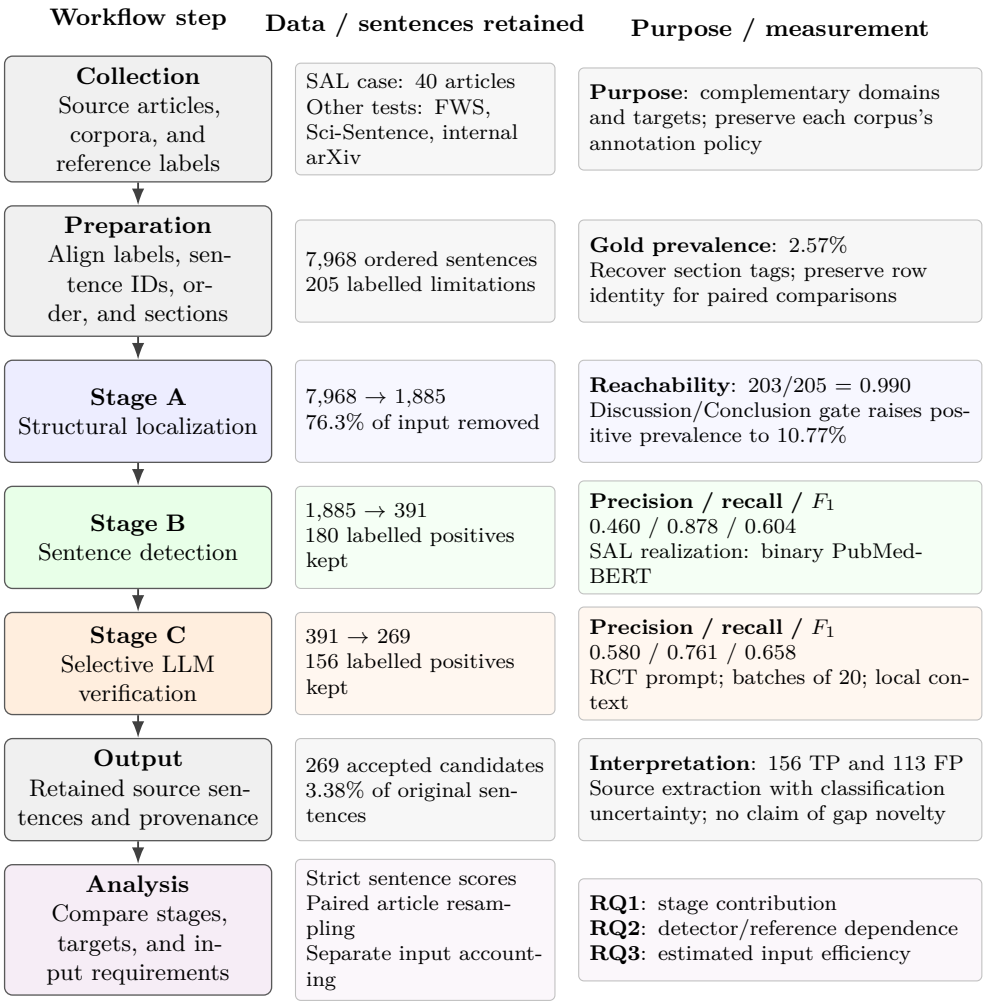


Figure 1: Workflow from collection and preparation to research-question analysis. Counts illustrate the saved SAL realization: recovered Discussion/Conclusion tags, a binary PubMedBERT detector, and RCT-prompt verification. Other experiments use the configurations in Sections 3.4 to 3.6; the hybrid runner and raw-PDF parsing are not evaluated by these SAL counts.

trained checkpoints are not included in the minimal branch. A new regenerator supplies the binary frozen-head recipe; exact checkpoint-level regeneration remains distinct from recomputing metrics from the saved predictions.

The internal 49-statement set contains scope, assumption, and open-problem labels in addition to the two detector classes. We use its target-specific limitation and future-work recalls only; recall over all 49 labels would have a different denominator and a different task definition. These labels were produced by an LLM and should be described as silver reference annotations. They neither establish human annotation agreement nor support precise system precision, because negative sentences were not exhaustively annotated. The original internal reference is separate from the older unarXive section-overlap benchmark.

3.3 Collection, preparation, and label alignment

The workflow begins by selecting a source corpus and recording its document identifiers, native annotation units, and intended target. This precedes model selection: a limitation-only biomed-

Table 2: Evaluation subsets and the surviving evidence used in the present study. "Aggregate" means that complete row-level predictions were not supplied for independent recomputation.

Subset	Size / target	Input condition	Evidence
SAL/RCT [Lan et al., 2024a]	40 articles; 7,968 sentences; 205 limitations	Reconstructed full-article stream; Discussion/Conclusion gate	Sentence labels, eight detector prediction vectors, and 391 verdicts for each of two prompts
SAL cost subset	10 articles; 2,053 sentences; 55 limitations	Sample restricted to articles with at least two positives	Stage counts and full-article aggregate metrics; token estimates over all 40 articles
FWS [Zhang et al., 2022]	1,506 sentences; 220 future-work positives	Curated future-work/conclusion material	Aggregate before/after filter metrics
Sci-Sentence [Bolaños et al., 2025]	140 test sentences; 40 Limitation/Research-Gap positives	Curated rhetorical sentences; no localization gate	Aggregate baseline/filter metrics; additional target-matched runs recorded in notes
Internal arXiv reference	10 articles; 49 mixed-type statements, including 11 limitations and 14 future-work statements	Parsed full articles; LLM-built silver labels	Notebook output and development summaries; original prediction files absent from the minimal snapshot

ical benchmark, a future-work sentence collection, and a rhetorical-role corpus cannot be made interchangeable merely by giving their positive labels the same name. The public resources supply complementary tests of the extraction components; the internal articles exercise parsing and transfer under a smaller, less reliable reference. We retain these distinctions when choosing eligible inputs and interpreting their outcomes.

For the SAL analysis, the collected material is the saved sentence stream and its recovered PMC section metadata. Article identifiers and sentence order connect the gold labels, section gate, detector predictions, and verifier verdicts. Preparation therefore preserves the row identities through every stage rather than independently resegmenting text for each model. This permits paired stage comparisons and prevents a removed sentence from disappearing from the recall denominator. The exact upstream reconstruction and any excluded source records must be recovered before rebuilding this stream from the original corpus; the present analysis starts from its saved 7,968 records.

FWS provides already curated sentences. Its breadth evaluation loads the available test material, excludes exact sentence matches to the training text, and selects the documented seeded subset. Sci-Sentence supplies a fixed test partition and rhetorical labels; Limitation and Research Gap are merged for its binary test. These preprocessing choices are explicit because curated sentences test recognition and verification, whereas reconstructed article streams additionally test localization. A train/test split should be tracked by article identifiers as well as text: exact sentence deduplication alone cannot exclude near duplicates or other sentences from the same article.

For raw articles, preparation instead extracts text blocks, identifies headings, segments sentences, and associates each sentence with its source section. The PDF libraries provide a local,

inspectable route that can retain layout cues without requiring an external parsing service. The resulting record must preserve enough source information to inspect errors in reading order, segmentation, and heading assignment. This route is implemented in the runner, but the SAL experiment's recovered tags do not demonstrate its accuracy. We distinguish an error caused by an absent candidate region from a detector error on an available sentence.

Label alignment is likewise recorded before scoring. Our principal definition excludes criticism of prior work unless the sentence also states a limitation or direction of the present study. For example, an authors' statement that their sample covers only one institution fits the limitation target; a claim that earlier methods ignore a phenomenon does not automatically fit it. A proposal to test a method on additional populations fits future work, while a description of an experiment already completed does not. These are illustrative policy examples, not additional annotated observations. Sci-Sentence's broader binary target is retained under its own definition; its Research Gap labels are not silently reannotated to match the author-only task.

3.4 Stage A: structural localization

The reusable runner invokes `slice_section_aware`. It unions the mid-paper-anchor localizer with complete sections whose headings name Discussion, Conclusion, Limitations, Future Work, or related expressions. The anchor localizer combines spans after target headings, inline keyword windows, the terminal tail, and narrow windows around scope/assumption phrases. Generic discussion anchors are position-gated; explicit limitation and future-work anchors are accepted independently of position. The terminal-window component caps selected units at 160; mid-paper additions and the whole-section union can expand that set, so the final localizer has no global 160-sentence cap. The resulting region list removes repeated source content according to the implementation's normalized-text keys.

For PDF input, `pdf_to_blocks` uses PyMuPDF, with pdfplumber as a fallback. Font size, bold weight, and short-line/numbering cues distinguish headings from body text. The heuristic uses a 1.12 body-font size ratio and a twelve-word heading limit. The sentence splitter protects common abbreviations and applies text normalization; references and selected terminal headings stop the eligible stream. These heuristics can introduce reading-order and sentence-boundary errors.

Structural localization is chosen as a low-cost, high-recall allocation mechanism. It reduces how many sentences the detector and verifier inspect while retaining regions where authors often qualify results or describe next steps. Whole-section retention protects statements without an explicit cue, while anchor windows support less conventional article structures. The choice has a direct measurable consequence: any labelled positive removed here is unreachable downstream. We therefore assess both compression and positive reachability rather than treating a smaller candidate pool alone as evidence of success.

The principal SAL experiment uses a simpler, explicitly distinct realization: retain sentences whose recovered section tag is `DISCUSS` or `CONCL`. This measures localization conditional on supplied section tags. It does not validate automatic heading recovery on PDFs. In the FWS curated pool the gate is effectively a pass-through; in Sci-Sentence it is disabled.

3.5 Stage B: sentence detection

The reusable hybrid detector combines lexical cues with a frozen sentence encoder and a class-weighted logistic regression. If a cue assigns a target label, that label is accepted; otherwise the highest-probability classifier label must be a gap type and meet threshold τ_B. The cleaned runner uses `BAAI/bge-small-en-v1.5`, $C = 2$, balanced class weights, and $\tau_B = 0.5$. Its expected training input is a cached self-distilled sentence set; development notes describe ACL limitation augmentation. The cache itself is absent from this snapshot. If neither a head nor its training data is available, the runner falls back to rules; that fallback must not be reported as a classifier run.

The final Stage-B notebook instead selects a frozen MiniLM-L6 encoder with logistic regression trained on Sci-Sentence, FWS, and SAL. These are separate configurations. Its training label map collapses Sci-Sentence Research Gap and Extension into the label named *future work* and Limitation into *limitation*. This is a proxy mapping: an Extension sentence is not necessarily future work. That semantic mismatch is a transfer limitation, not a change to the task definition. The development comparison also includes bge-base, e5-base, mpnet, and full fine-tuning of DistilBERT, SciBERT, and PubMedBERT [Sanh et al., 2019, Beltagy et al., 2019, Gu et al., 2021]. Frozen sentence features are obtained through Sentence-Transformers [Reimers and Gurevych, 2019].

Frozen encoders with a linear head separate semantic representation from task-specific fitting. They provide a comparatively small adaptation component and allow several representations to be compared without fine-tuning the entire encoder. Balanced class weights counter the dominance of non-target training sentences, and logistic regression supplies probabilities for thresholding. The regularization parameter and threshold specify the implemented operating point; they are not claimed to be universally optimal. Lexical cues offer an explicit path for conventional limitation and future-work expressions, but may confuse hypothetical statements, prior-work descriptions, and generic conclusions. The semantic head complements this path by recognizing wording beyond literal cue matches.

Fine-tuned scientific and biomedical encoders test a different adaptation strategy. Their inclusion asks whether representations specialized for scientific text and its domain improve detection on the evaluated stream. A detector ranking can reflect domain, training data, optimization, and threshold choice simultaneously. Accordingly, the saved-vector comparison answers which recorded configuration performed better; it does not isolate pretraining architecture as the sole cause.

In the SAL experiment, Stage B consists of eight saved binary detector prediction vectors: five frozen encoder heads and three fine-tuned encoders. Their names are retained in the result tables. The analysis applies Stage A to those predictions and compares them on the same source labels. The supplied `dump_sal_preds.py` regenerates the five frozen columns using normalized embeddings, balanced logistic regression ($C = 2$, maximum 2,000 iterations), SAL's training CSV, and a default positive-probability threshold of 0.5. It parses the list-valued `limitation_annotation` field and labels a sentence positive if any entry is true. It preserves the three fine-tuned columns because their checkpoints are not supplied. The notebook describes SAL-specific fitting and threshold tuning, but complete fine-tuning run manifests remain unavail-

able. Thus a specified frozen-head regeneration recipe does not establish bit-for-bit reproduction of every saved encoder or fine-tuned checkpoint.

3.6 Stage C: selective verification

The API verifier receives candidate sentences and returns structured `{id, keep}` judgments. It uses temperature zero, a strict JSON schema, and retries missing batch judgments through the single-item path. Empty single-item responses are handled conservatively by keeping the candidate; this is a failure policy, not a guarantee of semantic correctness. Surviving sentences are not rewritten.

The saved SAL run uses `yandexgpt-5-lite`, batches of 20 candidate sentences in document order, and at most one neighboring sentence on each side within the same article. It judges the actual PubMedBERT-positive stream rather than the union of positives from several detectors. Two cached prompt configurations are available: the keep-biased RCT prompt, labelled `rct` in the verdict file, and a keep-biased SAL-criteria prompt, labelled `sal`. Their complete target decisions are retained for all 391 candidates. The former depends on a historical prompt alias not defined in the minimal branch; the latter's wording is present in its experiment script. This provenance difference matters for regenerating API runs.

Verification is placed after detection because inspecting every full article with an LLM spends input on a large number of obvious negatives. Local context supplies attribution and temporal cues that an isolated candidate may omit: a preceding sentence can identify whose study is discussed, and a following sentence can clarify whether an action is proposed or already completed. Keeping context within the same article avoids attributing a candidate to an unrelated document. Batching amortizes repeated instructions, but may affect attention and judgments; a batch of 20 is the recorded configuration, not a demonstrated optimum.

The keep-biased instruction reflects the intended precision–recall trade-off. The verifier should remove clear off-target candidates while avoiding unnecessary deletion of valid but indirectly worded limitations. We therefore measure both false-positive removal and true-positive retention. A better final precision accompanied by a large recall loss is not described as an unqualified improvement. Structured identifiers connect decisions to their input rows and make the saved run checkable; a well-formed JSON reply itself does not validate the content of those decisions.

The implementation also contains general author-gap and future-work prompt modes. In the breadth experiment, FWS uses the future-work mode, whereas the initial Sci-Sentence run uses the author-gap mode despite its different attribution target. Target-matched Sci-Sentence prompt runs are treated as exploratory adaptation studies. The complete instruction is more than the selected base prompt. The shared batch wrapper appends an authors'-own limitation/assumption/scope/future-work decision rule even when the experiment overrides the base prompt. In particular, the intended gap-matched base prompt allows prior-work gaps while that suffix rejects them. We retain these recorded runs as exploratory configurations with conflicting instructions, rather than treating them as clean tests of target alignment. The appendix records both layers; silently repairing the wrapper would produce a new experiment, not reproduce the saved one.

Table 3: Operational design and justification for answering each research question.

RQ	Comparison / analysis	Reported measures	Why these measures fit
RQ1	Same SAL rows: full detector, A+B, then A+B+C; both saved prompts	Reachability, candidate counts, precision, recall, F_1, TP retention, FP removal	Separates where candidates are lost from whether verification improves the surviving set
RQ2	SAL detectors and classical baselines; auxiliary tasks; alternative references	Target-specific scores, candidate load, positional sensitivity, reference provenance	Separates detector behavior from label policy, reference coverage and test-pool differences
RQ3	Forty-paper input accounting; separate ten-paper full-article pilot	Summed estimated input tokens and ratio; pilot extraction metrics and failures	Includes prompt/context overhead and checks effectiveness alongside input reduction

The deployment runner protects selected cue matches in explicit gap sections; the direct SAL verdict experiment judges every binary-detector candidate. Batch size, context representation, judging order, and rule protection therefore belong to the experiment configuration, not to an assumed universal default. The command-line runner inherits a default batch chunk of 40 from `filter_gaps`; the direct SAL experiments explicitly use 20. The separate library entry point `extract_all_gaps` also restricts model-only acceptance by section, unlike the runner's simpler cue-or-head rule. We do not assign one implementation's measurements to every entry point.

3.7 Experimental design and comparisons

The design links each research question to a unit of analysis, a comparison, and a measurable answer (Table 3). The article is the sampling unit for uncertainty estimation, while sentence-level labels determine extraction correctness. Dataset-specific findings are kept separate so that differences in input preparation and annotation policy remain visible.

RQ1: stages on a common full-article stream. We recompute full-stream and A+B detector metrics from all 7,968 SAL records. We then apply the cached RCT and SAL-criteria judgments to PubMedBERT's A+B candidates. The recall denominator remains all 205 labelled limitations, including the two removed by Stage A. This prevents conditioning end-to-end recall only on reachable positives.

RQ2: detectors and annotation targets. The eight SAL detector vectors are compared under the same localization gate. Separate FWS and Sci-Sentence tables use the saved aggregate metrics and explicitly state their input condition. Development notes from the internal arXiv set report target-specific recalls and candidate load; these are not full prediction-level replications. The aggregate baseline grid includes an all-negative predictor, cue rules, TF-IDF with balanced logistic regression ($C = 4$), binary one-to-four-gram counts with Bernoulli naive Bayes, and normalized bge-small embeddings with balanced logistic regression ($C = 2$). Feature extraction is fitted on training material. SAL uses the saved full stream; FWS uses a separate seeded 20%-of-paper test split; Sci-Sentence uses its supplied test partition. The FWS baseline grid and 1,506-sentence breadth pool therefore have different test memberships and are not paired. A second

development notebook compares TF-IDF/SVM, training-only chi-squared feature selection with Bernoulli naive Bayes, frozen heads, and SciBERT fine-tuning on its own grouped splits. Those logged tables remain separate from the saved native-prevalence grid.

We do not pool the datasets or compare binary positive-class F_1 directly with a seven-class or macro-averaged published score.

RQ3: input accounting and a small full-article comparison. The accounting file estimates full-article and selective-verification input on all 40 SAL papers. It uses `cl100k_base` tokens, includes repeated prompts and surrounding context, and assumes per-paper batching at 20 candidates. Prompt/message wrappers contain approximate overhead constants. It is not a reconstruction of exact billed requests for the separately cached verifier stream. We report summed inputs and their ratio, excluding output tokens, embedding computation, model training, and API retries.

The full-article baseline sends numbered article sentences to the same Yandex model and extracts returned sentence indices. Its recorded run covers ten articles selected with seed 1 from papers having at least two labelled limitations. It uses a separate extraction prompt and does not implement chunking or a strong retry policy. Failed/unparseable replies yield no extracted indices. The recorded two empty/unparseable cases are part of that baseline's outcome, not removed from its reported recall denominator. The small sample is not representative of all papers, and the comparison does not isolate localization from prompt and output-format differences.

Selection and uncertainty. Prompts, detector choices, and thresholds were inspected during development, including feedback from benchmark subsets later used for reporting. The LLM comparisons are therefore exploratory. For the complete SAL prediction/verdict pair, we additionally perform 10,000 paired paper-cluster bootstrap resamples, with seed 20261013, recomputing pooled F_1 each time. Percentile intervals describe variability conditional on the saved predictions. They do not correct test-feedback selection, provider drift, or missing training provenance; no confirmatory significance or non-inferiority claim is made. Aggregate-only studies receive no invented confidence intervals or paired tests.

3.8 Measures and reproducibility

Strict sentence scoring is primary. Positive-class recall measures whether the system finds the annotated statements, while precision measures how much of the output a reader can accept under the reference policy. Their harmonic mean summarizes the trade-off without concealing the individual components. Accuracy would be dominated by the many negative sentences in the full SAL stream and is therefore not the headline measure. Candidate counts additionally reveal the work passed to subsequent stages, which a classification score alone does not capture.

Paper-cluster resampling preserves dependence among sentences from one article. Treating thousands of sentences as independent samples would overlook their shared writing style, topic, and section structure. Pairing the same resampled articles across stage configurations makes the difference analyzed comparable within each resample. Where only pooled metrics survive, we

report those metrics at their recorded precision and avoid constructing article-level uncertainty from them.

With TP, FP, and FN defined against the dataset's labels,

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2TP}{2TP + FP + FN}. \tag{2}$$

We report positive-class metrics and the native prevalence of each evaluated subset when that denominator is supplied. Development macro-F_1 values are labelled separately. Stage-A reachability is the proportion of labelled positives retained; compression is the proportion of all input sentences removed. Stage-C true-positive retention and false-positive removal are computed relative to its input detector and the *same gold labels*; they are not independent of annotation error.

For the internal parsed-PDF reference, legacy matching uses normalized token containment at 0.7. Token overlap tolerates boundary corruption but does not prove exact quotation or correct labeling. We additionally recompute the repository's neighboring-sentence scores for windows $W = 1, 2$ as a sensitivity analysis: a prediction within W positions of any gold sentence in the same paper contributes to precision, and a gold sentence within W positions of any prediction contributes to recall. This heuristic does not adjudicate semantic equivalence and can credit multiple nearby predictions. It is not substituted for strict scores or for manual review of disputed examples.

For prevalence sensitivity, we reuse the fixed SAL decisions inside the 1,885-sentence section pool. All 203 reachable positives are retained and 778 negatives are sampled without replacement, giving 981 sentences at approximately 20.7% prevalence. We average F_1 over seeds 0–7 using NumPy's default random generator. No new judging occurs after subsampling. This isolates the scoring effect of negative sampling conditional on the saved decisions; it is not a reproduction of the published authors' checkpoint or exact candidate pool.

3.9 Reference construction and the human-validation protocol

The branch also documents an expanded model-labelled reference with three positive types: limitation, future work, and research gap. The third label permits an unresolved issue in the field or prior literature; it does not add a third independently evaluated output class to the current two-target detector. The documented collection combines 190 arXiv CS/ML articles with the same 40 SAL articles. Separate whole-paper model passes propose sentence indices and short quotations. Quote checking is intended to verify source alignment. Agreement on the same type forms one tier; an additional model pass adjudicates one-sided or type-disagreement candidates to increase coverage. These tiers describe model agreement, not demonstrated human correctness.

Only the datasheet, the generated annotation interface, and its construction/merge utilities are supplied. The raw full-corpus labels, model responses, collection manifest, and generation scripts referenced by the datasheet are absent. Consequently, its corpus totals and inter-model agreement are reported as documented construction statistics, separate from prediction-level extraction results. Provider/model identifiers in that document have no accompanying request

logs or pinned versions; they are not evidence of a particular verified model release.

The implemented human-validation protocol samples four strata: 227 model-labelled limitation sentences not labelled in SAL, 15 SAL-positive disagreement items, 60 positive controls, and 60 negative controls. The generated interface contains 362 numbered targets from 40 articles; source/system labels are hidden, context is shown, and source articles are linked. Its builder uses seed 13. Two annotators can record a binary decision, a type, and notes. The merge utility computes coverage, raw agreement and Cohen's kappa on jointly answered rows; type agreement is conditional on both annotators accepting the item as a gap.

The utility also reports consensus-only acceptance proportions and a lenient either-annotator proportion. Disagreements are excluded from its consensus-only precision denominators; this is not an independent adjudication procedure. The stratum-enriched sample cannot establish population precision or full-corpus recall without appropriate sampling treatment and a reference for items missed by all models. It covers the biomedical disagreement problem, not independent validation of the 190 arXiv papers. No completed annotator exports or scored human results are supplied, so this protocol contributes no human-agreement or corrected-precision result in the present paper.

3.10 Artifact-based reproduction

The code snapshot examined is the `minimal` branch at commit `28a6c2d`. Our offline verification records hashes of all 72 tracked files, verifies prediction/verdict coverage and accounting totals, and generates the principal and supplementary numerical tables. No new detector training or LLM inference was performed for that recomputation. Original experiments used GPU fine-tuning and API-based judging; CPU-capable frozen heads are also implemented. Hard-coded machine paths, missing raw training inputs and checkpoints, and API aliases ending in `latest` currently limit exact end-to-end reproduction. The experiment prompts for SAL keep-biased and intended gap-matched validation are centralized in `stage_c_prompts.py`; shipped prompts remain in `gap_llm_filter.py`. The historical RCT alias is still imported from an absent helper. A changed detector stream requires new matching verdicts: retaining an old cache after candidate changes is not a valid pipeline rerun. The appendix distinguishes these regeneration constraints from the offline score recomputation, which requires neither training nor paid inference.

Data and artifact availability. The analyzed code and saved outputs are identified by the repository commit. Public benchmark sources are attributed in Table 2. The snapshot supplies the principal SAL predictions/verdicts and several aggregate results, but not all training inputs, model checkpoints, upstream reconstructions or expanded reference labels. Their absence limits end-to-end regeneration; it does not prevent reproducing the principal saved-run scores. Article text and benchmark data remain subject to their respective source licenses.

4 Results

The principal SAL tables are recomputed from saved row-level artifacts. The other tables distinguish aggregate result files from development notes. All values describe the recorded

Table 4: SAL stage contributions on 40 articles, with all 205 labelled positives in the recall denominator. TP denotes retained labelled positives. Both Stage-C rows use complete cached judgments on the same 391 PubMedBERT candidates.

Configuration	Kept	TP	Precision	Recall	F_1
Stage A	1885	203	0.108	0.990	0.194
A+B: PubMedBERT	391	180	0.460	0.878	0.604
A+B+C: RCT prompt	269	156	0.580	0.761	0.658
A+B+C: SAL keep-biased	249	148	0.594	0.722	0.652

configurations; none is treated as an untouched-test estimate of a universal final pipeline.

4.1 RQ1: localization, detection, and verification

Localizer development. The internal ten-paper notebook retains executed outputs for the terminal-window and mid-paper-anchor localizers. The former keeps 433 sentences and the latter 559, while reported localization at 0.70 token containment rises from 0.633 to 0.857. Drop rates are 87.3% and 83.6%. This reference includes scope and assumption statements as well as limitations and future work; these are region-coverage observations, not strict two-class extraction scores. At 0.90 containment, localization is lower (0.429 and 0.633), showing dependence on the overlap criterion. Full values and the logged section-aware comparison are in Section A.2.

The separate SAL-training development notebook reports five grouped-fold summaries of 0.861 mean reachability for anchor localization and 0.988 for whole target sections. The latter uses supplied section names; it does not measure PDF heading recovery. Its cue-augmented count adds potentially overlapping windows without full deduplication, so the archived augmented drop rates are approximate. These summaries describe a development protocol with token-containment scoring; they are not alternative estimates of the strict 40-paper evaluation below.

Fixed SAL test stream. Of 7,968 SAL sentences, 1,885 belong to the recovered Discussion or Conclusion sections. This gate discards 76.3% of the input while retaining 203 of 205 labelled limitations (reachability 0.990). Prevalence rises from 2.57% in the full stream to 10.77% in the retained pool. This is a conditional localization result using recovered tags rather than automatically parsed PDF headings.

Table 4 shows the resulting trade-off. PubMedBERT after localization retains 391 sentences, including 180 labelled positives. The RCT verifier removes 98 of the 211 detector false positives (46.4%), but also removes 24 of its 180 true positives. True-positive retention is therefore 86.7%. Precision increases from 0.460 to 0.580; recall decreases from 0.878 to 0.761; F_1 increases from 0.604 to 0.658. The SAL keep-biased alternative retains fewer sentences and achieves slightly higher precision, 0.594, but lower recall, 0.722, and F_1, 0.652.

The paired descriptive bootstrap interval for the RCT-prompt F_1 change is $[0.026, 0.083]$, around an observed increase of 0.054. The corresponding intervals for A+B and A+B+C are $[0.554, 0.654]$ and $[0.604, 0.710]$. These intervals concern the saved-run comparison only. Prompt refinement used benchmark feedback, so the positive interval does not establish a confirmatory generalization claim.

Table 5: SAL prevalence sensitivity with fixed predictions/verdicts. The section pool has 203 reachable positives; matching retains them and samples 778 negatives, averaging over eight seeds. Full-stream scores retain all 205 positives in the recall denominator.

Configuration	Full-stream F_1	Pool recall	Pool F_1	Matched F_1
A+B	0.604	0.887	0.606	0.749
A+B+C: RCT	0.658	0.768	0.661	0.757
A+B+C: SAL	0.652	0.729	0.655	0.741

Table 6: Eight saved SAL detector vectors scored on the full stream and after the same Discussion/Conclusion gate. These are positive-class metrics; the artifact permits metric recomputation but does not fully reconstruct each training job.

Stage-B detector	Full-stream F_1	A+B precision	A+B recall	A+B F_1
MiniLM-L6 (frozen)	0.216	0.249	0.727	0.371
mpnet (frozen)	0.280	0.304	0.776	0.437
bge-small (frozen)	0.282	0.298	0.810	0.436
bge-base (frozen)	0.294	0.299	0.815	0.437
e5-base (frozen)	0.314	0.312	0.849	0.457
DistilBERT-FT	0.413	0.377	0.849	0.523
SciBERT-FT	0.458	0.405	0.873	0.553
PubMedBERT-FT	0.495	0.460	0.878	0.604

An earlier notebook reports $F_1 = 0.701$ after judging a union of several detectors' candidates. That run and the candidate-specific cached run are not interchangeable, because batch composition and prompt configuration differ. We use the latter for the principal result. Nor is our full-stream F_1 directly comparable to the published 0.821 recognition result of Lan et al. [2024a], which used a different candidate-selection and model-development protocol.

The recomputed prevalence analysis makes this distinction explicit (Table 5). With negative subsampling to approximately 20.7%, detector F_1 is 0.749, RCT-verified F_1 is 0.757, and SAL-verified F_1 is 0.741. Thus the RCT verifier's advantage narrows, and the SAL variant no longer improves this measure. These are rescorings of fixed decisions, not new LLM runs. They demonstrate dependence on the evaluated negative pool, without establishing a universal prevalence threshold at which verification becomes useful.

Other prompt and model experiments have different judging populations. The stored 120-candidate balanced hard-sample sweep reports F_1 of 0.789, 0.779, and 0.784 for the three Yandex variants. Aggregate union-candidate runs include the strict SAL result of 0.568 and keep-biased SAL result of 0.554 for PubMedBERT; neither replaces its candidate-specific result of 0.652. Model, prompt and batch membership differ across some runs, preventing attribution of those differences solely to prompt wording. Section A.3 reports these development results separately.

4.2 RQ2: detector and reference dependence

Localization raises F_1 for every saved SAL detector vector (Table 6). PubMedBERT has the highest A+B F_1 in this comparison, followed by SciBERT and DistilBERT. The frozen heads retain useful recall but produce more false positives. This supports the localization mechanism for these fixed predictions; it does not establish that parameter count, pretraining domain, or

Table 7: Saved baseline-grid aggregates: positive-class metrics. SAL uses the 7,968-row full stream; FWS uses a separate paper-level 20% holdout, whose exact test count is not stored in the aggregate; SciSentence uses the supplied 140-row binary test.

Subset	Method	Precision	Recall	F_1
SAL	majority (all-neg)	0.000	0.000	0.000
SAL	TF-IDF + LogReg	0.362	0.571	0.443
SAL	CountBin(1-4) + BernoulliNB	0.000	0.000	0.000
SAL	cue-rules only	0.355	0.215	0.267
SAL	bge-small frozen + logreg	0.164	0.820	0.273
FWS	majority (all-neg)	0.000	0.000	0.000
FWS	TF-IDF + LogReg	0.667	0.767	0.714
FWS	CountBin(1-4) + BernoulliNB	0.880	0.510	0.646
FWS	cue-rules only	0.855	0.559	0.676
FWS	bge-small frozen + logreg	0.514	0.820	0.632
SciSentence	majority (all-neg)	0.000	0.000	0.000
SciSentence	TF-IDF + LogReg	0.784	0.725	0.753
SciSentence	CountBin(1-4) + BernoulliNB	0.895	0.425	0.576
SciSentence	cue-rules only	1.000	0.150	0.261
SciSentence	bge-small frozen + logreg	0.761	0.875	0.814

Table 8: Aggregate breadth results. FWS targets author-stated future work; Sci-Sentence merges Limitation and Research Gap and includes prior-work criticism. Neither row evaluates raw-PDF localization. The Sci-Sentence filter here uses the author-gap prompt and is deliberately reported as a target mismatch.

Subset	Configuration	Precision	Recall	F_1
FWS ($n = 1{,}506$)	A+B	0.723	0.618	0.667
	A+B+C, future-work prompt	0.843	0.609	0.707
Sci-Sentence ($n = 140$)	Detector	0.578	0.650	0.612
	Detector+C, author-gap prompt	0.900	0.450	0.600

fine-tuning alone caused the observed ranking.

Classical and frozen baselines. The saved native-prevalence baseline grid is reported in Table 7. On the full SAL stream, TF-IDF/logistic regression has $F_1 = 0.443$, compared with 0.273 for its separately fitted bge-small head; the cached bge-small column in Table 6 has 0.282. These are different saved fits, not rounding variants of one run. On the grid's FWS split, TF-IDF/logistic regression reaches 0.714, and on binary Sci-Sentence the in-domain bge-small head reaches 0.814. All-negative predictions have zero positive-class F_1. These results establish useful classical baselines and show why encoder size alone is an inadequate explanation of performance. The FWS grid score must not be treated as a paired comparison with the 0.707 breadth result, because their test pools differ.

The aggregate FWS run raises precision with little recorded recall loss, whereas the Sci-Sentence run raises precision at a larger recall cost and slightly reduces F_1 (Table 8). The latter's label definition permits prior-work gaps that an author-gap prompt rejects. Subsequent notes record target-matched prompt and in-domain threshold adaptation, but complete decisions for those additional runs are not available in the supplied snapshot. The development log reports $F_1 = 0.730$ for the intended gap-matched prompt with the existing cross-domain

Table 9: Internal arXiv development notes, restricted to the 11 limitation and 14 future-work reference statements. The 49-statement mixed-type denominator is not used. Retained-sentence counts indicate candidate load, not validated precision. No new training replication was performed.

Rules + detector	Limitation recall	Future-work recall	Retained sentences
bge-small (frozen; mixed-source training)	0.545	0.786	97
MiniLM-L6 (frozen; mixed-source training)	0.636	0.857	170
mpnet (frozen; mixed-source training)	0.455	0.786	186
PubMedBERT (fine-tuned; mixed-source training)	0.364	0.714	65

detector, and 0.880 for an in-domain bge-small head at threshold 0.3 followed by that prompt. Before verification, the latter retains 39 of 40 positives among 61 candidates; after verification it retains 33 positives among 35 candidates. These logged counts are arithmetically consistent with precision 0.943 and recall 0.825, but their full decisions are not supplied. The contradictory shared prompt suffix described in Section 3.6, together with test-informed development, limits interpretation as a clean policy-alignment experiment. Section A.4 records the configurations; none is an independent confirmation that the verifier is nearly error-free.

The internal reference illustrates a different operating-point trade-off (Table 9). MiniLM retains more target statements and more candidates than bge-small. The full fine-tuned model has lower recorded target recall on this small silver set. The result cannot establish that frozen encoders generally transfer better: the set is small, its reference coverage is incomplete, training-label mappings differ from the principal task, and precision was not independently measured. It also explains why the selected development notebook and the portable runner currently name different default heads. The complete logged head-only and hybrid comparisons, class-balancing checks, and within-corpus benchmark tables are retained in Section A.2. Validation macro-F_1 and positive-reference recovery answer different questions. Lower recovery after transfer is compatible with distribution shift, label-policy mismatch, incomplete silver references, or overfitting; these observations do not isolate one cause or establish a universal frozen-versus-fine-tuned ranking.

Alternative references further change the interpretation. The development notes describe a 33-statement two-target model reference and a stricter 19-statement reference. MiniLM's reported recovery changes from 0.455 to 0.632, while Stage-A reachability changes from 23/33 to 14/19. The latter reference was collected with incomplete reading of three long papers. This is sensitivity to annotation coverage and policy on the same articles, not a measured system improvement. The larger documented 230-paper reference and its pending human protocol are summarized in Section A.5. They supply no completed human accuracy result.

Neighbor-position scoring on the saved SAL stream also changes the numbers. For A+B, F_1 rises from 0.604 under strict scoring to 0.743 at $W = 1$; RCT-verified output rises from 0.658 to 0.821. The latter number is coincidentally equal, after rounding, to the published SAL detector score under a different protocol. It is not evidence of matching that system. Neighbor scoring can credit multiple predictions around one gold sentence and does not establish semantic correctness or repaired annotations (Table 20).

Table 10: Estimated input volume over the 40 SAL articles. The accounting includes repeated verification prompts and local context, using `cl100k_base` and approximate wrapper constants. It excludes output tokens, inference latency, local detector work, and retries.

Input accounting	Total input tokens	Mean per article
Full-article prompting	282,846	7,071.2
Selective verification	75,622	1,890.6

Table 11: Recorded ten-article SAL comparison: 2,053 sentences and 55 positives, sampled from articles having at least two labelled limitations. The complete stage-count file survives; full-article extraction survives as aggregate precision/recall and mean API input only. The baseline F_1 is approximately 0.427, computed from its rounded precision and recall.

Configuration	Kept	Precision	Recall	F_1
Stage A	498	0.110	1.000	0.199
A+B: PubMedBERT	109	0.468	0.927	0.622
A+B+C: RCT verifier	77	0.584	0.818	0.682
Full-article LLM	—	0.559	0.345	≈0.427

4.3 RQ3: input volume and full-article prompting

The accounting file gives a ratio of 282,846/75,622 = 3.740, a 73.3% reduction in estimated LLM input (Table 10). Only 391 of 7,968 source sentences reach verification, about 9.8 candidates per article. The token ratio is smaller than the sentence-count ratio because repeated prompts and neighboring context contribute substantial overhead. It is not a measured monetary saving, throughput increase, or constant per-article cost.

In this selected sample, A+B+C has higher recorded recall and F_1 than the full-article baseline using the same model family (Table 11). The baseline's mean reported API input is 6,540 tokens per article, while the 40-article accounting table concerns a different sample and an estimated rather than billed input measure. We do not combine those two measures into a sample-specific cost ratio. The baseline also lacks a robust chunking/retry protocol; two replies were recorded as empty or unparseable. Consequently, the comparison includes both semantic extraction and response-format robustness and is not evidence that a well-engineered full-article LLM baseline, or a stronger model, would have the same performance.

5 Discussion

5.1 What the stage comparison establishes

The principal finding is a measured allocation trade-off. On the fixed SAL prediction artifact, structural localization removes many non-target sentences with little loss of labelled positives. The detector then reduces candidate load, and selective verification improves precision while sacrificing some of the detector's recall. The RCT-prompt F_1 increase of 0.054 is useful for this configuration, but its 13.3% loss of detector true positives is also material. Verification is therefore an optional operating-point choice, not a universally beneficial final step.

The result depends on where a stage is evaluated. A full-article detector sees a low-prevalence

stream, whereas curated conclusion sentences or a balanced recognition sample present a different false-positive burden. Changing the negative sample changes precision and F_1 even when positive recall is unchanged. Published values must consequently be matched by label definition, candidate pool, prevalence, and metric before being treated as head-to-head comparisons.

5.2 How the findings answer the research questions

For RQ1, the stage counts explain both the gain and its cost. Localization retains almost all labelled positives, whereas most of the final recall loss occurs in detection and verification. A practitioner prioritizing discovery can retain the detector's broader candidate set; a reader prioritizing a shorter review list may prefer the verified output. Both outputs preserve the original sentence and can be exposed with their decision provenance. The experiment supports this operating-point distinction rather than a single mandatory setting for every application.

For RQ2, the recorded SAL comparison favors the fine-tuned biomedical detector among the eight saved vectors, while the separate full-stream baseline grid also shows a strong lexical baseline. The breadth results caution against assuming that its ranking or verifier policy transfers unchanged. Future-work recognition, own-study limitation recognition, and broader research-gap recognition require different judgments. A reusable system should therefore expose the target policy and its corresponding evaluation. The conflicting suffix in the current gap-matched experiment also shows that policy must be checked in the assembled request, not just in the base prompt. The present evidence is strongest for the saved SAL configuration and weaker for aggregate-only transfer studies and silver-label internal observations.

For RQ3, selective processing provides a clear reduction in estimated verifier input under the specified accounting profile. This result explains why the staged design is attractive when many articles must be inspected. It does not determine the best overall deployment architecture: a small local detector adds its own preparation and maintenance costs, and a hosted model's response reliability affects both effectiveness and retries. These quantities should be measured when making a system-level efficiency claim. The branch's projected cross-paper batching, prompt caching and long-paper savings are scenarios rather than measured deployment improvements; we do not promote them to billed-cost or throughput results.

5.3 Attribution policy and reference labels

The Sci-Sentence development experiments illustrate the importance of matching the intended prompt policy to the reference task, although the conflicting shared wrapper prevents a clean causal test of that change. A verifier instructed to reject criticism of prior studies can correctly follow that instruction while disagreeing with a dataset that labels those criticisms as Research Gap or Limitation. This is a target mismatch, not evidence that either definition is intrinsically wrong. Likewise, a statement about a study's methodological constraint and an adjacent consequence can be scored differently under sentence- or span-oriented guidelines.

Inspection of disagreements can reveal annotation ambiguities, missed labels, model errors, and sentence-boundary errors. It cannot by itself establish that most false positives are correct or that an LLM achieves a much higher "true" precision than its measured precision. Such a claim requires independent, preferably domain-expert, adjudication of a representative disagreement

sample. We therefore retain the original benchmark labels in all reported metrics. Neighbor-credit scores, if used, should be interpreted as positional sensitivity rather than corrected gold.

Source grounding has similar limits. Returning an existing parsed sentence avoids inventing the sentence's wording, but does not establish that it is a genuine author-stated gap, that its subject is the authors' own work, or that the gap remains unresolved. Future downstream systems would need separate attribution, resolution, and novelty checks before using these extractions to propose research.

5.4 Efficiency and deployment

Selecting candidate sentences substantially reduces estimated LLM input in the measured SAL accounting profile. The 3.74-fold ratio incorporates some repeated-prompt and context overhead, which makes it more informative than simply quoting the percentage of sentences removed. However, a complete system cost also includes parsing, embeddings, detector training and inference, response retries, and output tokens. API billing depends on the actual tokenizer and provider pricing. Prompt caching and cross-paper batching could change that profile, but their projected savings are not measured deployment results here.

The ten-article full-LLM comparison is encouraging but limited. It uses one economical model, one selected sample, a different extraction prompt, and a baseline with response-format failures. A stronger comparison would hold the annotation policy fixed, save all responses, use equivalent retry/chunking budgets, and measure both stage-wise effectiveness and actual resource use. The present result should guide that evaluation rather than substitute for it.

5.5 Implications for evidence-based research support

An extracted statement is useful as a traceable starting point for literature analysis. It can help readers locate a paper's qualifications, collect proposed follow-up studies, and compare how a topic is framed across articles. Keeping the source wording and identifier allows those readers to inspect the surrounding argument rather than relying on an isolated model summary. However, aggregation requires further decisions: two sentences can express the same issue, a proposed study may already have been completed, and a frequently mentioned limitation may reflect reporting convention rather than scientific priority.

Broader human evaluation and venue-corpus analysis can address some of these questions. Such analyses must connect their article samples, annotation definitions, and run configurations to the extraction protocol. Annotation agreement, venue-level statement frequencies, and the usefulness of generated research ideas each require their own measurements. They cannot be inferred from a sentence-detection score or a reduction in verifier input alone.

5.6 Threats to validity

Selection and sampling. The verifier prompts were refined after inspecting benchmark feedback. Descriptive bootstrap intervals capture variation among the saved articles, not the optimism introduced by that selection. The next evaluation should freeze prompts, head, threshold, context, and batching before examining an independent article set. The cost subset is addition-

ally conditioned on at least two labelled limitations per article; its outcomes need not describe papers with few or no such statements. The FWS breadth selection excludes exact training sentences but does not certify exclusion of all related or same-paper training material.

Measurement and transfer. Recovered BioC tags simplify full-article localization relative to noisy PDFs. Curated FWS and Sci-Sentence inputs do not test that front end at all. The internal arXiv references and the larger documented reference are model-labelled. The 362-row interface has no completed human results and samples biomedical disagreements, so it cannot establish reference accuracy across both domains. Inter-model kappa is not a substitute for human agreement. The small arXiv reference is silver, positive-only, and contains an expanded label ontology; it supports limited target-specific recall observations, not population accuracy or human agreement. Proxy training-label mappings, including Extension to future work, further limit interpretation of cross-domain rankings.

Reproducibility. Principal SAL decisions survive and their scores can be checked without an API, but the fine-tuned detectors still lack checkpoints and full training provenance; the newly supplied frozen-head regenerator resolves only part of that gap. Some experiment scripts retain machine-specific paths or imports of helpers absent from the minimal branch. Experiment base prompts have been centralized, but the historic RCT alias still needs to be archived with its exact resolved wording. Models referenced through a provider's `latest` alias may change, and temperature zero does not establish repeatability across versions or batch compositions. Aggregate-only breadth and full-article results cannot support additional paired significance analyses or error adjudication without their original predictions and responses.

These limits bound the current conclusions: a particular saved full-article funnel improves its recorded precision and F_1, and selective processing reduces estimated input volume. Independent human-annotated results, broader venue findings, and evaluated research-idea quality are outside the evidence reported in this study.

6 Conclusion

We studied an extractive funnel that localizes candidate regions, detects author-stated gap sentences, and optionally verifies candidates with an LLM. In the reconstructed 40-article SAL stream, localization retains 203 of 205 labelled limitations while discarding 76.3% of sentences. For the saved PubMedBERT predictions, selective verification raises precision from 0.460 to 0.580 and F_1 from 0.604 to 0.658, at a recall decrease from 0.878 to 0.761. These results answer RQ1 for the recorded configuration rather than for an independently frozen test protocol.

For RQ2, classical baselines, detector-development studies and auxiliary tasks show that candidate load, domain, annotation target and reference coverage affect the operating point. Group-partition localization summaries, alternative silver references and positional sensitivity scores support narrower conclusions than a single pooled accuracy figure. A literature-gap label policy must not be silently replaced by an authors'-own limitation policy. For RQ3, accounting on the 40 articles gives a 3.74-fold reduction in estimated LLM input; the selected ten-article

baseline study records higher funnel recall and F_1, but does not establish a general advantage over all full-article LLM methods.

The branch additionally supplies a model-labelled reference-construction account and a blind human-validation interface. Their existence does not establish completed human validation. The shared prompt-wrapper conflict should be resolved and the revised request frozen before new inference is evaluated. The next step is an independent evaluation with frozen configurations, complete training and prediction provenance, adjudicated errors, and measured total processing cost. Reliable source extraction may support later idea-generation systems, but their novelty and usefulness require separate experiments.

A Configurations and supplementary experiments

The manuscript concerns the `minimal` branch at `28a6c2d`. Source-level experiment coverage is provided in the accompanying `EXPERIMENT_COVERAGE.md`. Supplementary tables preserve completed development comparisons without elevating notebook prose or aggregate-only runs to the evidential status of saved sentence decisions. All comparisons retain their own task, denominator, sampling procedure, and scoring convention.

A.1 Implementation configurations and reproduction

Table 12: Distinct configurations in the branch. A shared stage decomposition does not imply identical detector training or API requests.

Configuration	Localization	Detection	Verification
Command-line runner	Anchors union whole target sections; local PDF blocks	bge-small, cached three-class head; cues or threshold 0.5	General/RCT base; protected rules; default chunk 40
Final Stage-B notebook	Cached internal candidate regions	MiniLM, mixed Sci-Sentence/FWS/SAL; cues or head argmax	Not part of that notebook's metric
SAL primary test	Recovered Discussion/Conclusion labels	Eight saved binary vectors; PubMedBERT supplies 391 candidates	Two cached prompts, chunk 20, same-paper neighboring context, every candidate judged
FWS breadth	Curated sentences; pass-through	Production-style bge hybrid	Future-work mode, chunk 20
Sci-Sentence adaptation	No section gate	Cross-domain hybrid or in-domain binary bge head	Intended gap override with conflicting shared suffix

The offline verifier checks all 391 candidate IDs against both cached verdict maps, recomputes strict scores, paired bootstrap intervals, negative-subsampling sensitivity and positional scores, and sums the token-accounting CSV. Source hashes and the complete tracked-file manifest identify the analyzed snapshot. No training or API calls are required for these computations. The new frozen-head regenerator specifies the five binary encoder-head recipes but requires the external SAL training CSV. Fine-tuned predictions remain cached until their original checkpoints are supplied. Both frozen-head regeneration and new judging are separate from score reproduction.

The principal SAL prediction fixture and its two complete verdict maps support offline scoring. Separate files store the baseline-grid, breadth, token-accounting and ten-article pilot results. Model and prompt sweeps preserve additional aggregate comparisons. Internal transfer and public-benchmark development values survive in notebook reference tables and the Stage-B result summaries; their referenced prediction/checkpoint files are not supplied. The companion experiment inventory maps each result to its exact repository path and manuscript location.

A.2 Localization and detector development

Table 13: Executed internal localizer outputs: ten articles, 49 mixed-type silver statements. Localization is token containment within the concatenated selected region text, not one-to-one sentence matching.

Variant	Kept	Drop (%)	Loc@0.70	Loc@0.80	Loc@0.90
Terminal windows (v1)	433	87.3	0.633	0.551	0.429
Mid-paper anchors (v2)	559	83.6	0.857	0.735	0.633

The notebook's narrative states that section-aware v3 matches v2 on the fragmented internal blocks; no independently executed v3 metric row survives. The final-notebook PDF front end is implemented and demonstrated, but no exhaustive PDF parsing/heading accuracy study is supplied. GROBID section slicing is also implemented as an alternative; its presence is not an evaluated GROBID-versus-local-parser comparison.

Table 14: Notebook reference values for SAL-training localization development. Values are means and standard deviations across five article-grouped partitions. Reachability uses token containment at 0.70. Cue-augmented drop accounting can count overlapping windows more than once.

Selection rule	Reachability	Reported drop (%)
Anchor localizer	0.861 ± 0.054	82.2
Anchors + curated cues	0.880 ± 0.047	79.5
Whole target sections	0.988 ± 0.020	75.8
Whole sections + cues (logged)	0.990 ± 0.020	75.4

The same development notebook reports single-split anchor reachability of 0.861/0.766/0.935 for train/validation/test and 0.882/0.798/0.946 after curated cues. It attributes the remaining misses mainly to cue-less Discussion statements. These observations motivate whole-section retention; their fold variability is descriptive, not a significance test. The archived cue list is fixed across the grouped-fold summaries rather than re-mined inside every fold.

Table 15: Logged detector development on the internal reference. SF means Sci-Sentence+FWS; SFS adds SAL. Validation is pooled macro-F_1; L/F are limitation/future-work recovery against 11/14 silver reference statements. Head and hybrid columns share the cached localizer input. Retained counts do not measure precision.

Detector	Train	Val macro	Head L	Head F	Hybrid L	Hybrid F	Kept
bge-small	SF	0.633	0.273	0.714	0.455	0.786	104
DistilBERT-FT	SF	0.745	0.000	0.643	0.182	0.714	63
SciBERT-FT	SF	0.846	0.000	0.643	0.273	0.714	61
bge-small	SFS	0.718	0.364	0.786	0.545	0.786	97
PubMedBERT-FT	SFS	0.890	0.182	0.714	0.364	0.714	65
bge-base	SFS	0.742	0.273	0.786	0.455	0.786	76
e5-base	SFS	0.753	0.273	0.786	0.455	0.786	71
mpnet	SFS	0.668	0.364	0.714	0.455	0.786	186
MiniLM-L6	SFS	0.656	0.364	0.714	0.636	0.857	170

The original SF training report describes 6,700 rows, including only 101 limitations, and full fine-tuning with AdamW, learning rate 2×10^{-5}, three epochs and class-weighted loss. The later mixed-source report lists 7,706 training rows. These are development mixtures, not the command-line runner's unavailable self-distilled training cache. The larger frozen representations have 768-dimensional embeddings; bge-small and MiniLM use 384 dimensions. Reported validation macro-F_1 increases under full fine-tuning do not imply improved transfer. The retained notes disagree on some per-dataset macro conventions; we use the consistent pooled values and target-specific internal recoveries rather than mixing incompatible columns.

The MiniLM balancing notebook reports mixed-ontology recovery of 0.571 with balanced class weights, 0.265 without weights, 0.551 with explicit resampling, and 0.571 with additional paraphrases. The denominator is all 49 statements, including non-target types, and the per-class decisions do not survive. These numbers document a development choice; they are not two-target accuracy estimates or evidence that weighting is universally optimal.

Cue optimization is also recorded separately. On a paper-disjoint FWS development test, the reported precision/recall/F_1 changes from 0.843/0.448/0.585 to 0.819/0.538/0.650 with curated intent cues. Limitation-cue F_1 on SAL changes only from 0.267 to 0.275. A more permissive future-keyword variant was rejected after development feedback and is not used as a held-out performance claim. Its selection history further motivates caution about generalization.

Table 16: Logged within-corpus benchmark development, separate from the saved baseline grid. FWS is a balanced paper-grouped split, SAL samples about four negatives per positive from training articles before a new split, and Sci-Sentence collapses seven labels into three proxy classes on a new grouped split. Macro and positive/type F_1 are distinct measures.

Task	Method	Macro-F_1	Positive/type F_1
FWS binary	Cues	0.676	0.585
	bge-small	0.836	0.833
	MiniLM	0.791	0.787
	TF-IDF/SVM	0.835	0.833
	Counts/chi-squared/NB	0.837	0.829
	SciBERT-FT	0.876	0.880
SAL binary	Cues	0.581	0.256
	bge-small	0.813	0.710
	MiniLM	0.729	0.584
	TF-IDF/SVM	0.833	0.735
	Counts/chi-squared/NB	0.834	0.729
	SciBERT-FT	0.900	0.843
Sci-Sentence proxy	bge-small	0.714	L 0.744; F 0.643
	MiniLM	0.622	L 0.667; F 0.517
	TF-IDF/SVM	0.652	L 0.458; F 0.684
	Counts/chi-squared/NB	0.358	L 0.087; F 0.356
	SciBERT-FT	0.856	L 0.706; F 0.916

These notebook reference tables do not establish superiority to the original papers' results: the split, negative sampling, class maps and sometimes the averaging convention differ. Earlier logged FWS recognition values (including 0.879 for bge-small) concern another run; they are not substituted for the 0.836 macro-F_1 table above. Claims that an original FWS result is inflated by leakage cannot be independently established from this branch's missing reproduction outputs. Regularized fine-tuning and SPECTER2 are described as unrun or failed, so they have no performance row. The older ACL-cap/self-distillation sequence is described in a package README but not supported by completed outputs in the current all-experiments notebook.

Alternative model-built references on the internal articles contain 33 target statements (40,000-character full-text cap) or 19 target statements in nine articles (9 limitations, 10 future-work statements). On the former, MiniLM recovery is 0.455 overall, 0.500 for limitation and 0.421 for future work, with 23/33 reachable. On the latter, MiniLM and bge-small hybrid recovery is 0.632, with per-type values 0.444/0.800 and 14/19 reachable; head-only recovery is 0.579. Their retained loads are 167/102 for the hybrids and 157/91 for the heads. Incomplete reading of three long papers and differing annotation policies preclude treating either reference as exhaustive human gold.

A.3 Verifier model, prompt and judging-population experiments

Table 17: Saved balanced hard-sample sweep: 120 detector candidates, 60 positive and 60 negative, SAL keep-biased prompt, batches of 20. These selected-candidate results are not full-stream extraction scores. Calls are the recorded counts, not dollar cost or latency.

Judge	Precision	Recall	F_1	Calls
yandexgpt-5-lite	0.683	0.933	0.789	6
yandexgpt-5-pro	0.697	0.883	0.779	6
yandexgpt-5.1	0.645	1.000	0.784	6

An earlier notebook implements a five-model by three-batch probe on 30 positive and 30 negative SAL sentences (batch sizes 10, 20 and 40), but supplies no executed numerical table. Its qualitative selection of the lite model at batch 20 is therefore a documented development decision, not an independently checked optimum. A later context/batch/prompt diagnosis logs a best hard-sample F_1 near 0.827 without full decisions or a recoverable configuration table. The development summary also reports that 56/60 gold positives were kept in one mixed batch setting versus 37/60 in a denser stream. These observations motivate candidate-specific judging, but the missing batch-level records prevent a controlled causal estimate of the batch effect.

Table 18: Saved aggregate union-candidate runs. Strict SAL uses a different recorded judge/run from the lite keep-biased variants; columns are not a controlled prompt-only ablation. Each verifier run judged the union of 1,036 detector candidates before scoring a detector's subset.

Detector	A+B F_1	Strict SAL	SAL keep-biased	RCT
MiniLM-L6 (frozen)	0.371	0.490	0.445	0.435
mpnet (frozen)	0.437	0.519	0.471	0.472
bge-small (frozen)	0.436	0.533	0.484	0.482
bge-base (frozen)	0.437	0.527	0.484	0.465
e5-base (frozen)	0.457	0.536	0.499	0.490
DistilBERT-FT	0.523	0.556	0.544	0.537
SciBERT-FT	0.553	0.562	0.545	0.547
PubMedBERT-FT	0.604	0.568	0.554	0.567

An earlier broad-prompt union table survives only as notebook reference values. Its eight A+B+C F_1 values are 0.576 (MiniLM), 0.623 (mpnet), 0.616 (bge-small), 0.620 (bge-base), 0.648 (e5-base), 0.691 (DistilBERT), 0.686 (SciBERT), and 0.701 (PubMedBERT). The candidate-specific RCT result of 0.658 supersedes the latter as the principal deployment-stream estimate; differences also involve run configuration. A verdict is not detector-independent when the model receives several candidates jointly. The saved aggregate union runs demonstrate that verification can also lower F_1: PubMedBERT's 0.604 A+B becomes 0.554 or 0.567 in the two keep-biased union runs.

A single-reader analysis in the development summary categorizes 22 of the SAL keep-biased run's 101 false positives, suggesting a mixture of missed labels, policy differences, and model errors. This is useful for hypothesis formation, but the sampling rule and independent adjudication are unavailable. We neither replace the benchmark labels nor report the summary's projected corrected precision.

The Kilicoglu cross-check is likewise a notebook-only report: 1,505 test sentences, approximately 31.8% positive, bge/logistic detection at P/R/F_1 0.719/0.781/0.749 and subsequent verification at 0.876/0.503/0.639. No row predictions or implemented reproduction of the original authors' rule-based baseline are supplied. This recorded precision increase and recall loss does not establish widespread gold-label errors or evaluate the full PDF front end.

A.4 Breadth adaptation and scoring sensitivity

Table 19: Logged Sci-Sentence adaptation, all on the 140-sentence binary test. The base prompt is intended to allow prior-work gaps but retains an incompatible authors'-own wrapper suffix. Threshold/prompt selection was exploratory; complete row decisions are absent.

Configuration	Precision	Recall	F_1
Cross-domain hybrid + intended gap override	1.000	0.575	0.730
In-domain binary bge, threshold 0.5	0.761	0.875	0.814
In-domain binary bge, threshold 0.3	0.639	0.975	0.772
Same threshold 0.3 + intended gap override	0.943	0.825	0.880

The in-domain Sci-Sentence script explores thresholds 0.5, 0.4, 0.3 and 0.2 and can judge the last two settings. Only the reported settings above have supplied numerical summaries. The analogous FWS recall-first script is implemented, with a different article split and threshold sweep, but no completed output is supplied. Neither script's mere presence supplies additional measured operating points. FWS disagreement examples are hypotheses about label coverage, not adjudicated evidence of reference error.

Table 20: Recomputed positional sensitivity on the fixed SAL decisions. Neighbor credit uses many-to-many positional matching within each article; it is not a semantic span annotation or an adjudicated correction of the benchmark.

Configuration	Strict F_1	Neighbor $W = 1$	Neighbor $W = 2$
A+B	0.604	0.743	0.783
A+B+C: RCT	0.658	0.821	0.853
A+B+C: SAL	0.652	0.817	0.852

A.5 Documented reference construction and pending validation

Table 21: Construction totals reported in the branch's annotation datasheet. These values are arithmetically consistent but not recomputed from the unavailable corpus/vote files. They are model-labelled reference statistics, not independently validated extraction results.

Item	Documented value
Articles / sentences	230 / 68,719 (190 arXiv articles and the same 40 SAL articles)
Positive statements	4,462: 2,234 research gaps, 1,543 limitations, 685 future-work statements
Positive statements by domain	3,751 arXiv; 711 SAL
Model-agreement / adjudicated tiers	2,960 / 1,502
Adjudication candidates	1,737; 1,502 accepted and 235 rejected
Inter-model Cohen's kappa	Any gap 0.778; limitation 0.748; future work 0.869; research gap 0.725
SAL reference recovery	Any-gap recall 0.956; limitation-specific recall 0.927 (190/205)
SAL limitation extras	227; a further model pass accepts 116/227 (51.1%)
Reported annotation expenditure	Approximately USD 37.5, including labeling, adjudication and the additional model pass
Human study	362-row interface present; completed answers/results absent

The reported model strings are `claude-opus-5-5`, `claude-sonnet-5-5`, and `claude-fable-5-1` with an Opus fallback. They are transcribed provenance identifiers from the datasheet, not independently verified provider versions. No request/usage records support exact rerunning or billing reconstruction. The 227 extras belong to this reference construction, not to the 113 or 101 false positives of the two saved funnel configurations. Likewise, limitation-specific recovery and any-gap recovery have different positive definitions.

Inter-model agreement must not be equated with human agreement or a human performance ceiling. Another LLM's acceptance of 116 extras cannot determine that the remaining items are errors, nor that the accepted ones are definitively correct. Quote alignment verifies text identity, not label semantics. The planned human protocol in Section 3.9 is an appropriate next check, but supplies no completed validation result here. Its biomedical, disagreement-enriched sample would not by itself validate every type and domain in the documented corpus.

A.6 Exact surviving instructions and composition

The verifier's batch schema returns a `results` array of integer `id` and Boolean `keep`. Missing batch decisions fall back to a single-target call, which loses the supplied neighbor context; an empty single-target response is kept conservatively. The current module centralizes two experiment base prompts. The complete request additionally contains a shared wrapper, calibration examples and numbered candidates. The following instruction is the surviving SAL keep-biased base, followed by its examples.

SAL keep-biased system instruction.

> You judge whether the «marked» TARGET sentence is a SELF-ACKNOWLEDGED LIMITATION of the AUTHORS' OWN study, using the SAL annotation criteria. A limitation is a weakness/caveat/bias/constraint the authors admit about THIS trial, fitting a known

category: study design; loose or narrow population / convenience sampling; single-center or narrow setting; intervention composite/non-standardized; control (no placebo / active placebo / care-as-usual); outcome measure excluded/imprecise/unvalidated; missing data or drop-out; small sample / underpowered; randomization unbalanced/poor/open; blinding (patient or team not blinded); short duration or follow-up; multiple/post-hoc testing or unadjusted confounding; funding; limited generalizability.

DECISION RULE (keep-biased):

- keep=true if the TARGET states such a limitation, OR states a consequence ('could introduce bias', 'cannot be assessed', 'results may be confounded', 'risk of type 1 error') whose underlying CAUSE (no blinding, small sample, single center, multiple testing, short follow-up...) is evident in the TARGET or its SURROUNDING context. When in doubt and a real weakness of this study is present, keep=true.

- keep=false ONLY for CLEAR non-limitations: restated positive results; clinical recommendations; critique of prior/other work; plain methods description; background; or a bare header like 'This study has several limitations.' with no specific weakness.

Few-shot pairs.

YES: the lack of a no treatment arm in this study

YES: our study included only a relatively small sample of participants

YES: The absence of masking is most relevant for the assessment of events.

YES: the crossover design and the relatively short treatment periods are limitations

YES: the trial was conducted at a single center

YES: Because outcomes were exploratory, the risk of type 1 error was considerable.

YES: This could have introduced a bias in the results.

NO: This study has several limitations.

NO: The intervention significantly reduced symptoms at 12 weeks.

NO: Clinicians should consider these findings when counseling patients.

NO: Existing methods cannot capture long-range dependencies.

NO: Participants were randomized 1:1 using a computer-generated sequence.

Intended gap-matched base instruction.

You are a precision filter for RESEARCH-GAP and LIMITATION statements in academic text, including related-work sections. KEEP the «marked» sentence if it identifies a limitation, weakness, shortcoming, unsolved problem, open challenge, or missing capability — in prior/existing work, a method, or the field (it need NOT be the authors' own study). DROP it if it merely describes what a work does, gives background, states results/findings, or makes a neutral or positive general statement with no gap or weakness. When a sentence clearly names a shortcoming or an unaddressed problem, KEEP.

Calibration pairs. YES: Existing methods cannot capture long-range dependencies.
YES: However, these approaches do not scale to large graphs.
YES: No prior work has addressed the multilingual setting.

YES: This remains an open problem in the field.
YES: Current benchmarks lack annotations for rare classes.
NO: archjava is an architecture description language on top of java.
NO: The model achieves 92% accuracy on the benchmark.
NO: Smith et al. proposed a transformer-based model for this task.
NO: These tools are widely used in industry.

Shared decision suffix (verbatim).

> You will receive a NUMBERED list. For EACH item, decide keep=true if the TARGET states the AUTHORS' OWN limitation, assumption, scope restriction, or future-work direction, and keep=false otherwise (including limitations of prior or other work).

This suffix conflicts with the gap-matched base's permission for prior-work criticism. Recording it is necessary to interpret the saved results faithfully. The historical RCT-loose alias is still imported from an absent helper and should not be replaced silently by the shipped RCT instruction. Centralizing the SAL/GAP bases improves future maintainability but does not recover that missing historical request.

A.7 Evidence outside the reported experiments

Historical LimGen binary-detection, BAGELS coverage and alternative FWS/SAL leaderboard values appear in comparison notes without their producing scripts, checkpoints or predictions in this snapshot. They establish development history, not independently checkable benchmark superiority. The corresponding inventory entries record their values and exclusion reasons. Proposed cross-provider LLM baselines, regularized fine-tuning and idea-generation evaluation have no completed supporting results in this branch. Project plans are not experiment outcomes.

The previously discussed 64-article human study and 800-article venue corpus have no supplied results in this snapshot. They are not replaced by the newer 362-row validation interface or the documented 230-paper model-labelled collection: these are distinct samples and protocols. No human agreement, venue distribution, temporal trend, or idea-generation quality is inferred from their descriptions.

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